



NATIONAL UNIVERSITY OF MODERN LANGUAGES

Artificial Intelligence Lab Report No - 1

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PROGRAM: BSSE (6th Semester)

COURSE: Artificial Intelligence

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AI Lab Task – Python String Operations

Objective

To learn basic Python string operations including:

1. Printing a statement
2. Counting characters
3. Converting to uppercase
4. Converting to lowercase
5. Printing characters one by one
6. Printing a statement in reverse

Introduction

In this lab task, we used Python programming in Google Colab to perform different operations on strings. Strings are sequences of characters and are widely used in Artificial Intelligence for text processing and data handling.

Lab Task Code

1. Print a Statement

```
[1]  print("hello World")  
✓ Os  
▼ ... hello World
```

2. Count Number of Characters

```
[2]  text = "Hi how are you"  
✓ Os  
count = len(text)  
print("Total number of characters:", count)  
▼ ... Total number of characters: 14
```

3. Convert to Uppercase

```
[3]  text = "hi gee how are u"  
✓ Os  
upper_text = text.upper()  
print("Uppercase:", upper_text)  
▼ ... Uppercase: HI GEE HOW ARE U
```

4. Convert to Lowercase

```
[5] text = "HI AGAIN NOW MY TEXT IS UPPERCASE"

lower_text = text.lower()
print("Lowercase:", lower_text)
```

... Lowercase: hi again now my text is uppercase

5. Print Characters One by One

```
[6] text = "Rafay"

for character in text:
    print(character)
```

... R
a
f
a
y

6. Print Statement in Reverse

```
[7] text = "M Abdul Rafay Khan"

reverse_text = text[::-1]
print("Reversed Statement:", reverse_text)
```

... Reversed Statement: nahk yafar luda M

Explanation

- `print()` → Displays output on the screen.
- `len()` → Counts total characters in a string.
- `.upper()` → Converts text into uppercase letters.
- `.lower()` → Converts text into lowercase letters.
- `for` loop → Prints each character separately.
- `[::-1]` → Reverses the string using slicing.